

PRACTICAL 5

Title

Apply Array and Object Handling – Create a Cart Total Calculator with Discount Logic

Aim

To develop a simple **Cart Total Calculator** using HTML, CSS, and JavaScript by applying **arrays and objects** to store product information and implementing **discount logic** to calculate the final cart amount.

Objectives

- To understand the use of arrays in JavaScript.
- To understand and manipulate objects containing product details.
- To calculate individual product subtotals.
- To calculate the total cart value.
- To apply a discount based on the cart total.
- To display the final payable amount dynamically on a webpage.

Files Required

1. index.html
2. style.css
3. script.js

1. index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Cart Total Calculator</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="container">
    <h1>Cart Total Calculator</h1>

    <div id="cart"></div>
```

```
        <div class="summary">
          <p>Subtotal: ₹<span id="subtotal">0.00</span></p>
          <p>Discount: ₹<span id="discount">0.00</span></p>
          <h2>Final Total: ₹<span id="total">0.00</span></h2>
        </div>

        <button onclick="calculateCart()">Calculate Total</button>
      </div>

      <script src="script.js"></script>
    </body>
  </html>
```

2. style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
}

.container {
  width: 500px;
  background-color: white;
  padding: 30px;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.15);
}

h1 {
  text-align: center;
  color: #333;
  margin-bottom: 25px;
}

.cart-item {
  display: flex;
  justify-content: space-between;
  padding: 12px;
  margin-bottom: 10px;
  background-color: #f8f8f8;
  border-radius: 6px;
}

.summary {
  margin-top: 20px;
  padding: 15px;
  background-color: #eef5ff;
```

```
        border-radius: 8px;
    }

    .summary p,
    .summary h2 {
        margin-bottom: 10px;
    }

    button {
        width: 100%;
        padding: 12px;
        margin-top: 20px;
        border: none;
        border-radius: 6px;
        background-color: #007bff;
        color: white;
        font-size: 16px;
        cursor: pointer;
    }

    button:hover {
        background-color: #0056b3;
    }
```

3. script.js

```
// Array containing product objects
const cart = [
    {
        name: "Laptop",
        price: 50000,
        quantity: 1
    },
    {
        name: "Mouse",
        price: 800,
        quantity: 2
    },
    {
        name: "Keyboard",
        price: 1500,
        quantity: 1
    },
    {
        name: "Headphones",
        price: 2000,
        quantity: 1
    }
];

// Display cart items
function displayCart() {
    const cartContainer = document.getElementById("cart");

    cartContainer.innerHTML = "";

    cart.forEach(function(item) {
        const itemTotal = item.price * item.quantity;
```

```

        cartContainer.innerHTML += `
            <div class="cart-item">
                <span>
                    ${item.name} × ${item.quantity}
                </span>

                <span>
                    ₹${itemTotal.toFixed(2)}
                </span>
            </div>
        `;
    });
}

// Calculate cart total
function calculateCart() {

    // Calculate subtotal using array and objects
    const subtotal = cart.reduce(function(total, item) {
        return total + (item.price * item.quantity);
    }, 0);

    let discount = 0;

    // Discount logic
    if (subtotal >= 50000) {
        discount = subtotal * 0.10;    // 10% discount
    }
    else if (subtotal >= 20000) {
        discount = subtotal * 0.05;    // 5% discount
    }
    else {
        discount = 0;                  // No discount
    }

    const finalTotal = subtotal - discount;

    // Display calculated values
    document.getElementById("subtotal").textContent =
        subtotal.toFixed(2);

    document.getElementById("discount").textContent =
        discount.toFixed(2);

    document.getElementById("total").textContent =
        finalTotal.toFixed(2);
}

// Display cart when page loads
displayCart();

```

Working / Explanation

The program uses an **array of objects** to store the products in the shopping cart.

Each object contains three properties:

- name – stores the product name.
- price – stores the price of the product.
- quantity – stores the quantity purchased.

For example:

```
{
  name: "Laptop",
  price: 50000,
  quantity: 1
}
```

The `reduce()` method is used to calculate the subtotal of all products:

```
const subtotal = cart.reduce(function(total, item) {
  return total + (item.price * item.quantity);
}, 0);
```

The program then applies discount logic:

```
Cart value ≥ ₹50,000    → 10% discount
Cart value ≥ ₹20,000    → 5% discount
Cart value < ₹20,000    → No discount
```

The final amount is calculated using:

```
finalTotal = subtotal - discount;
```

Sample Calculation

For the given cart:

Product	Price	Quantity	Amount
Laptop	₹50,000	1	₹50,000
Mouse	₹800	2	₹1,600
Keyboard	₹1,500	1	₹1,500
Headphones	₹2,000	1	₹2,000
Subtotal			₹55,100

Since the subtotal is greater than ₹50,000, a **10% discount** is applied.

```
Discount = ₹55,100 × 10%
          = ₹5,510
```

```
Final Total = ₹55,100 - ₹5,510
            = ₹49,590
```

Result

The Cart Total Calculator was successfully created using **HTML, CSS, and JavaScript**. The program demonstrates **array and object handling**, calculates product subtotals, applies discount conditions, and displays the final payable cart amount dynamically.

Conclusion

This practical demonstrates how JavaScript arrays and objects can be used to represent real-world shopping-cart data. The `reduce()` method helps calculate the cart subtotal efficiently, while conditional statements implement the required discount logic.